

YIBO PENG

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EDUCATION

Carnegie Mellon University

Master of Science in Artificial Intelligence Engineering

Pittsburgh, PA

Aug 2023 – Dec 2024

Beijing Jiaotong University & Lancaster University

Bachelor of Science in Computer Science (Honours)

Beijing, CN & Lancaster, UK

Aug 2018 – July 2022

PUBLICATIONS

- [1] P. Xia*, **Y. Peng***, J. Wang*, K. Zeng, X. Wu, X. Tang, H. Zhu, Y. Li, S. Liu, Y. Lu, H. Yao. “MMedAgent-RL: Optimizing Multi-Agent Collaboration for Multimodal Medical Reasoning,” in Proceedings of the 14th International Conference on Learning Representations (**ICLR 2026**), Rio de Janeiro, Brazil, April 2026. [Paper]
- [2] **Y. Peng***, J. Song*, L. Li*, X. Yang, M. Christodorescu, R. Mangal, C. Pasareanu, H. Zheng, B. Chen. “When *Correct* Is Not Safe: Can We Trust Functionally Correct Patches Generated by Code Agents?” in Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**), San Diego, California, July 2026. [Paper]
- [3] **Y. Peng***, P. Xia*, D. Zhong*, K. Zeng, S. Han, Y. Zhou, J. Liu, R. Zhang, H. Yao. “SimpleOCR: Rendering Visualized Questions to Teach MLLMs to Read.” in Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**), San Diego, California, July 2026. [Paper]
- [4] **Y. Peng**, Z. Wang, D. Fried. “Can Long-Context Language Models Solve Repository-Level Code Generation?” *LTI Student Research Symposium*, 2025 (Poster). [Paper]
- [5] **Y. Peng**, S. Chen, L. Lian, D. Wagner. “SecOPD: Mitigating Adaptive Prompt Injections by On-Policy Distillation.” *Submitted to the Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*.
- [6] S. Chen, **Y. Peng**, C. Sitawarin, D. Wagner. “SecPO: Principled Adversarial Training for Prompt Injection Security.” *Submitted to the Conference on Neural Information Processing Systems (NeurIPS 2026)*.

RESEARCH EXPERIENCE

Berkeley Security Lab, University of California, Berkeley

Researcher

Feb 2026 – Present

- Researched secure LLMs and model-level defenses against prompt injection attacks
- Researched adversarial post-training and preference optimization for LLM security
- Evaluated prompt-injection robustness for instruction-following and agentic tool-calling systems
- Advisor: [David Wagner](#)

InfiniAI Lab, Carnegie Mellon University

Researcher

May 2025 – Present

- Researched security and reliability risks in LLM-based code agents
- Researched vulnerability injection in functionally correct code patches
- Built evaluation pipelines for repository-level code repair and code-agent robustness
- Advisor: [Beidi Chen](#)

AIMINGLAB, UNC Chapel Hill

Researcher

Oct 2025 – Jan 2026

- Researched multimodal language models and visual text understanding
- Researched OCR-oriented training methods for improving visual grounding
- Evaluated multimodal reasoning models on OCR-intensive benchmarks

- Advisor: [Huaxiu Yao](#)

Microsoft Research *Researcher*

Feb 2025 – May 2025

- Researched multi-agent systems for multimodal medical reasoning
- Researched reinforcement learning methods for coordinating medical reasoning agents
- Evaluated multimodal medical agents on visual question answering benchmarks
- Advisor: [Yan Lu & Huaxiu Yao](#)

Language Technologies Institute, Carnegie Mellon University *Researcher* *Jan 2025 – May 2025*

- Researched code LLMs for repository-level code generation and code understanding
- Researched retrieval-augmented generation and long-context modeling for coding tasks
- Built RAG-based semantic code search tools for LLM code agents
- Advisor: [Daniel Fried & Graham Neubig](#)

WORK EXPERIENCE

PricewaterhouseCoopers LLP (PwC)

Beijing, CN

Development Engineer Intern – Quantitative Model Expert Team

Nov 2021 – Apr 2022

- Developed a large VBA application to assess and calculate Expected Credit Loss (ECL) of accounts receivable.
- Reduced calculation time from **15 minutes to 10 seconds** by transitioning calculations to the database.
- Improved code efficiency by simplifying loops, reducing global variable usage, and optimizing function calls.
- Collaborated with cross-functional teams to integrate the model and over **230 listed companies** used it.

SERVICES

Reviewer: ACM Computing Surveys, NeurIPS 2026, CVPR 2026, NeurIPS 2025 Efficient Reasoning Workshop; NeurIPS 2025 ResponsibleFM Workshop; ICML 2025 R2-FM Workshop